

SYSTEM MANAGEMENT ADMINISTRATION

Project Plan

Project Title:

Small Office Network Design with Security Implementation

I. Objectives

- 1. To design a functional small office network
- 2. To identify key network components (server, clients, firewall)
- 3. To implement basic network security practices
- 4. To create a structured and secure network layout
- 5. To document and explain the system design clearly

II. Network Overview

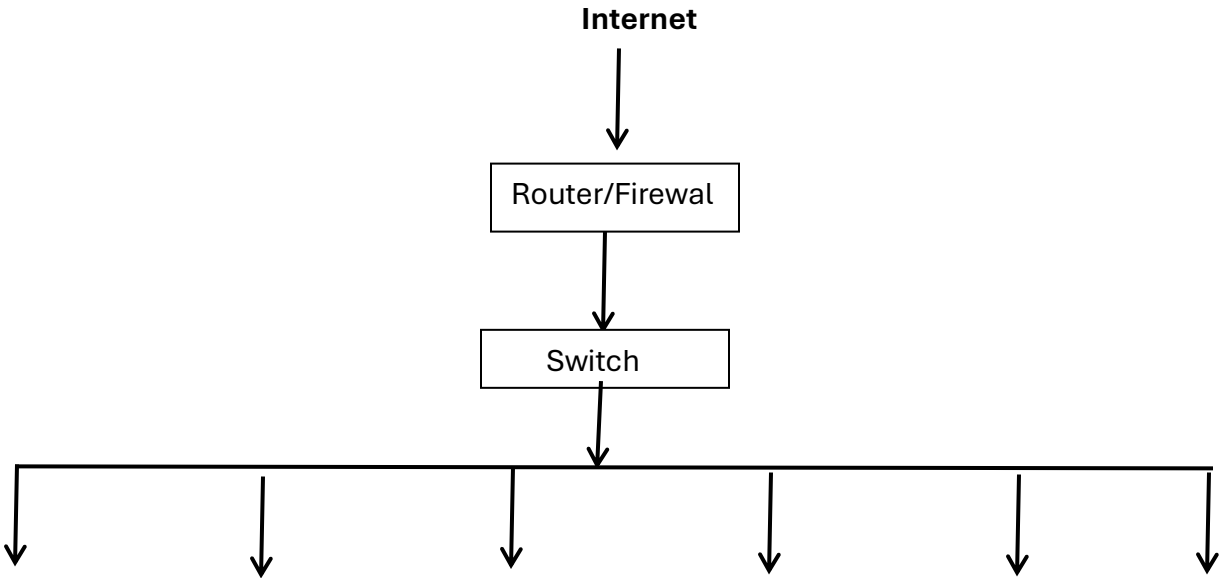
This project presents a network design for a small office with 6–10 users. The system includes:

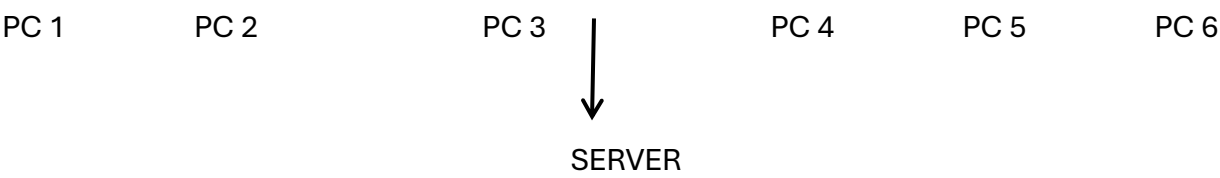
- 1 Server
- Multiple Client Computers
- Firewall (Router-based)
- Internet Connection

III. Network Components

| Component       | Description                                   |
|-----------------|-----------------------------------------------|
| Router/Firewall | Controls traffic and provides internet access |
| Server          | Stores files and manages users                |
| Client PCs      | Used by employees                             |
| Switch          | Connects all devices in the network           |
| Internet        | External network access                       |

IV. Network Diagram Optional





V. IP Addressing Table ( This is for Network configuration only) Optional

| Device IP | Address       | Description |
|-----------|---------------|-------------|
| Router    | 192.168.1.1   | Gateway     |
| Server    | 192.168.1.10  | File Server |
| Client1   | 192.168.1.101 | Employee PC |
| Client2   | 192.168.1.102 | Employee PC |
| Client3   | 192.168.1.103 | Employee PC |

VI. Server Configuration Optional

| Service             | Purpose             |
|---------------------|---------------------|
| File Sharing        | Store company files |
| User Authentication | Login control       |
| Backup System       | Data protection     |

VII. Security Plan Optional

1. Firewall Rules

| Rule                  | Action     | Description            |
|-----------------------|------------|------------------------|
| Allow HTTP/HTTPS      | Allow      | Web browsing           |
| Allow SSH             | Restricted | Admin access only      |
| Block Unknown Traffic | Deny       | Protection on Security |

2. Security Measures Optional

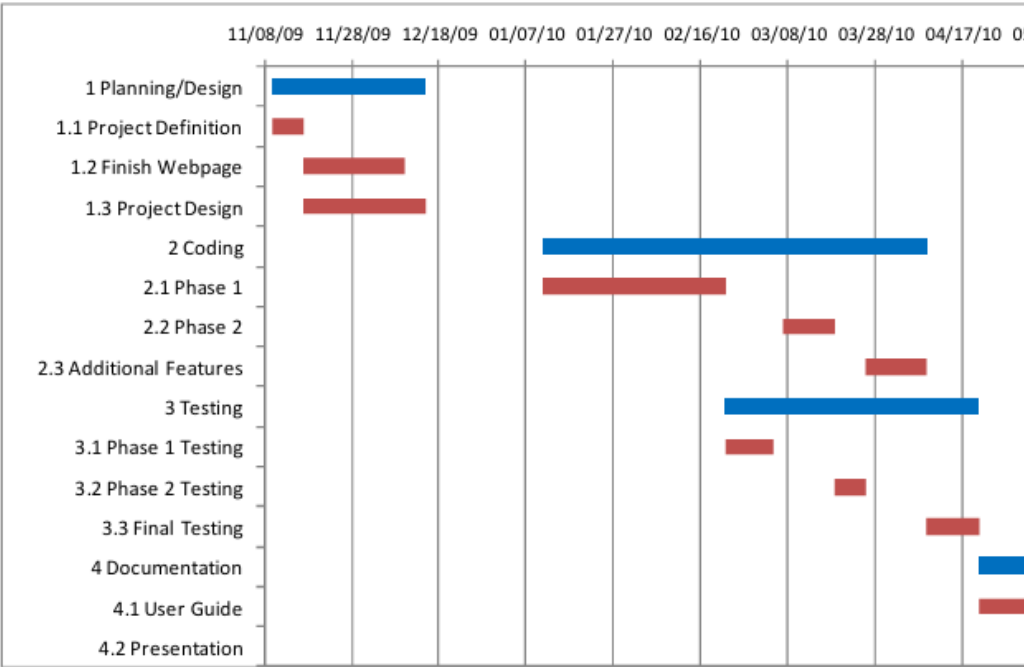
- Strong passwords for all users
- Regular system updates

Antivirus installation  
Limited user access (role-based)  
Backup every week

VIII. Schedule

The Gantt Chart below is a tentative schedule of our plans for the remainder of the current semester and next semester. We will be using this schedule to make sure we stay on focus, however; plans are not set completely and therefore might be changed before spring semester begins.

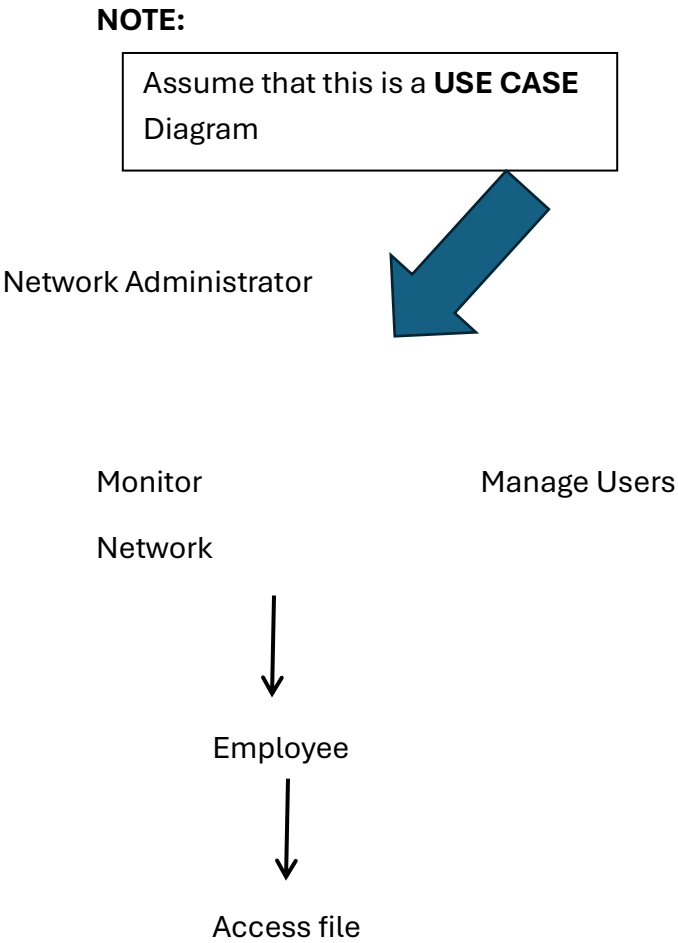
| Computer Science Gradebook Gannt Chart |            |                |            |
|----------------------------------------|------------|----------------|------------|
| Tasks                                  | Start Date | Duration(Days) | End Date   |
| 1 Planning/Design                      | 11/10/2009 | 35             | 12/15/2009 |
| 1.1 Project Definition                 | 11/10/2009 | 7              | 11/17/2009 |
| 1.2 Finish Webpage                     | 11/17/2009 | 23             | 12/10/2009 |
| 1.3 Project Design                     | 11/17/2009 | 28             | 12/15/2009 |
| 2 Coding                               | 1/11/2010  | 88             | 4/9/2010   |
| 2.1 Phase 1                            | 1/11/2010  | 42             | 2/22/2010  |
| 2.2 Phase 2                            | 3/7/2010   | 12             | 3/19/2010  |
| 2.3 Additional Features                | 3/26/2010  | 14             | 4/9/2010   |
| 3 Testing                              | 2/22/2010  | 58             | 4/21/2010  |
| 3.1 Phase 1 Testing                    | 2/22/2010  | 11             | 3/5/2010   |
| 3.2 Phase 2 Testing                    | 3/19/2010  | 7              | 3/26/2010  |
| 3.3 Final Testing                      | 4/9/2010   | 12             | 4/21/2010  |
| 4 Documentation                        | 4/21/2010  | 16             | 5/7/2010   |
| 4.1 User Guide                         | 4/21/2010  | 12             | 5/3/2010   |
| 4.2 Presentation                       | 5/3/2010   | 4              | 5/7/2010   |



IX. Use Case Diagram

Actors:

- Network Administrator
  - Employee
- Use Cases:**
- Access Files
  - Manage Network
  - Configure Security
  - Monitor System



**X. Budgetary Management Plan**

This section outlines the estimated cost needed to implement the proposed small office network

**A. Hardware Cost Estimate**

| Item                     | Quantity | Unit Cost (PHP) | Total Cost (PHP) |
|--------------------------|----------|-----------------|------------------|
| Router/Firewall          | 1        | 3,000           | 3,000            |
| Network Switch (16-port) | 1        | 2,500           | 2,500            |
| Server Computer          | 1        | 35,000          | 35,000           |
| Client Computers         | 5        | 20,000          | 100,000          |
| Ethernet Cables          | 10       | 150             | 1,500            |
| UPS (Backup Power)       | 1        | 3,000           | 3,000            |

Sub Total: 145, 000.00

| Software           | Cost (PHP) |
|--------------------|------------|
| Ubuntu Server      | Free       |
| Antivirus Software | 2,000      |
| Office Software    | 5,000      |
|                    |            |
|                    |            |

Sub Total: 7,000.00

Miscellaneous Costs

| Item  | Cost  |
|-------|-------|
| _____ | _____ |
| _____ | _____ |

TOTAL ESTIMATED BUDGET: PHP \_\_\_\_\_

XI. Project Staffing (Development Phase)

In this area, describes the personnel involved in the creation and development of the project.

A. Project Developer

| Name     | Role                                    | Responsibilities                                                                          |
|----------|-----------------------------------------|-------------------------------------------------------------------------------------------|
| Mr. Bean | System/Network Designer & Administrator | Designed the network, configured system, created documentation, implemented security plan |
| _____    | _____                                   | _____                                                                                     |
| _____    | _____                                   | _____                                                                                     |
| _____    | _____                                   | _____                                                                                     |
| _____    | _____                                   | _____                                                                                     |

B. Roles Performed by the Developer

| Role             | Description                                  |
|------------------|----------------------------------------------|
| Network Designer | Created the network layout and IP addressing |

|                      |                                           |
|----------------------|-------------------------------------------|
| System Administrator | Configured server, users, and security    |
| Security Analyst     | Designed firewall rules and security plan |
| Technical Writer     | Prepared documentation and reports        |

**XII. Write your Conclusion / Reflection.**

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Members:

Instructor :